

National Institute of Technology Kurukshetra

Department of Mechanical Engineering

Mid Sem-I (Feb. 2026)

B. Tech. (Mechanical) 4<sup>th</sup> Semester

Time: 50 Min.

Total marks: 15

Sub.: Fluid Machinery (MEPC210)

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*Instructions: 1) All questions are compulsory.*

*2) Assume suitable data, if required, and mention it.*

*3) Solve the numerical stepwise, indicating SI units.*

*4) Avoid writing unnecessary/unasked information in the answer.*

1. The peripheral velocity at the inlet of an inward flow reaction turbine is 30 m/s. The velocity of whirl at the inlet is 21.6 m/s, and the velocity of flow at the inlet is 3 m/s. Discharge of the turbine is 1180 liters/s. Hydraulic efficiency is 80%. Determine the turbine head, the runner power, and the inlet vane angle. Assume that the turbine is discharging radially at the exit.
2. A Pelton wheel having a wheel diameter of 1.2 m rotates at 600 rpm. Water is supplied at the rate of  $0.6 \text{ m}^3/\text{s}$  under a head of 450 m. If the buckets deflect the jet by an angle of  $160^\circ$ , find the power developed and the turbine's hydraulic efficiency. Assume a coefficient of velocity of 0.97 and neglect frictional losses in the bucket.
3. Define the specific speed of a hydraulic turbine. Classify the turbines based on specific speed. Also, describe the functions of penstock, surge tank, draft tube, guide vane and spiral casing with a schematic layout.

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*All the Best*